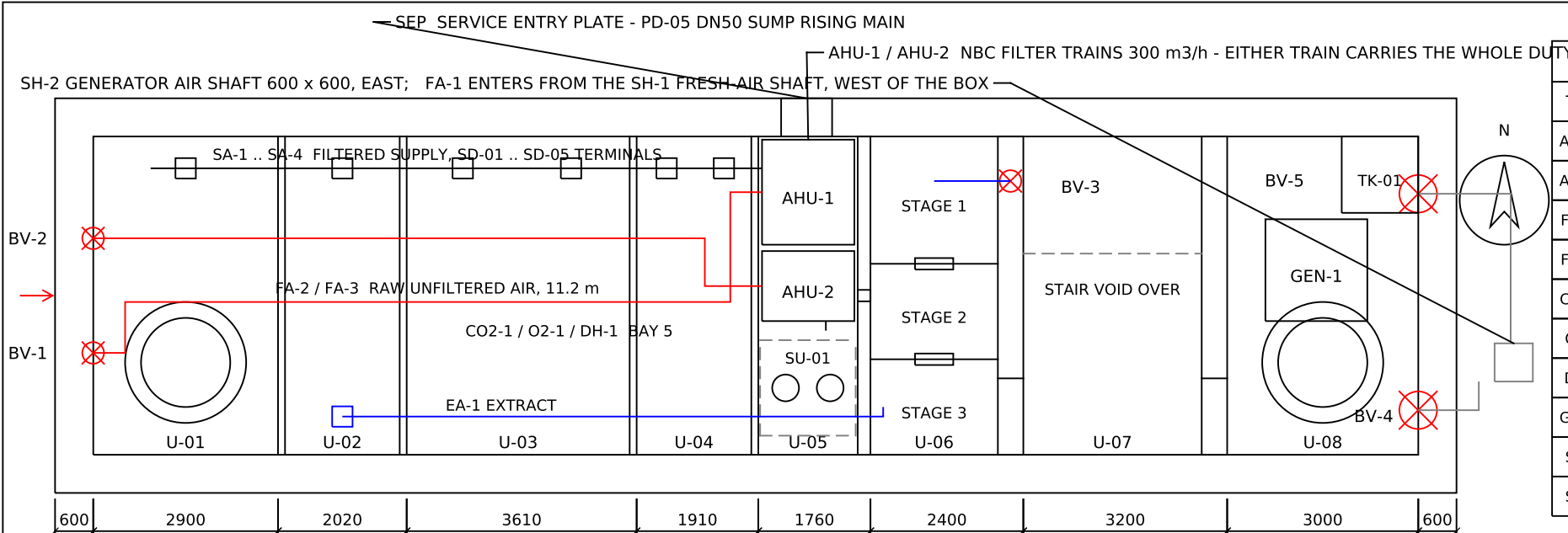
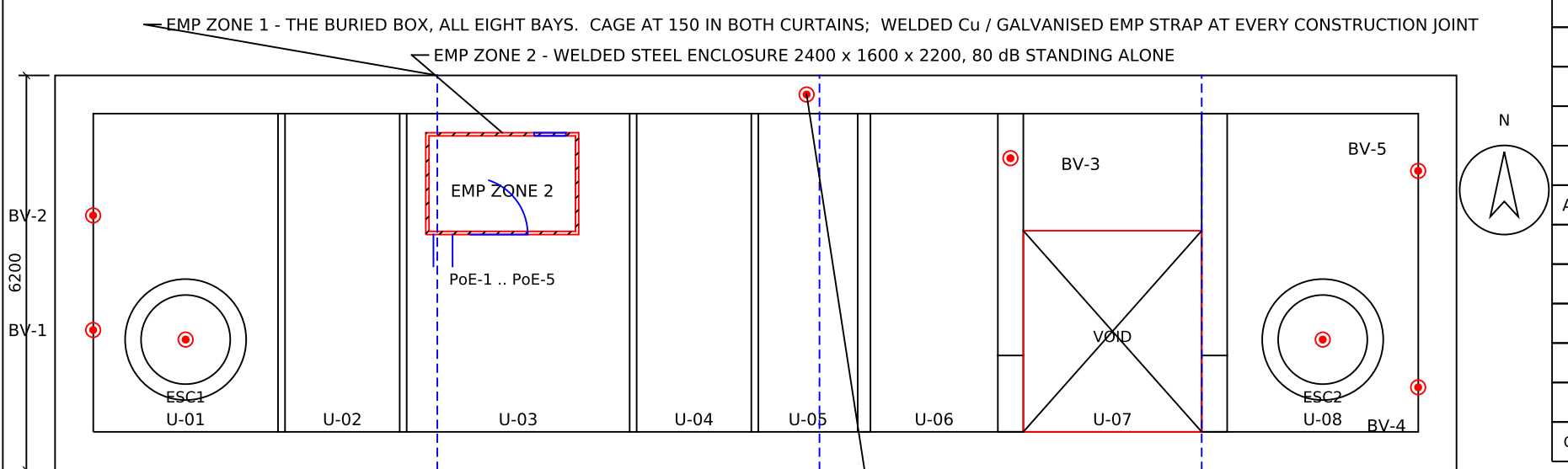
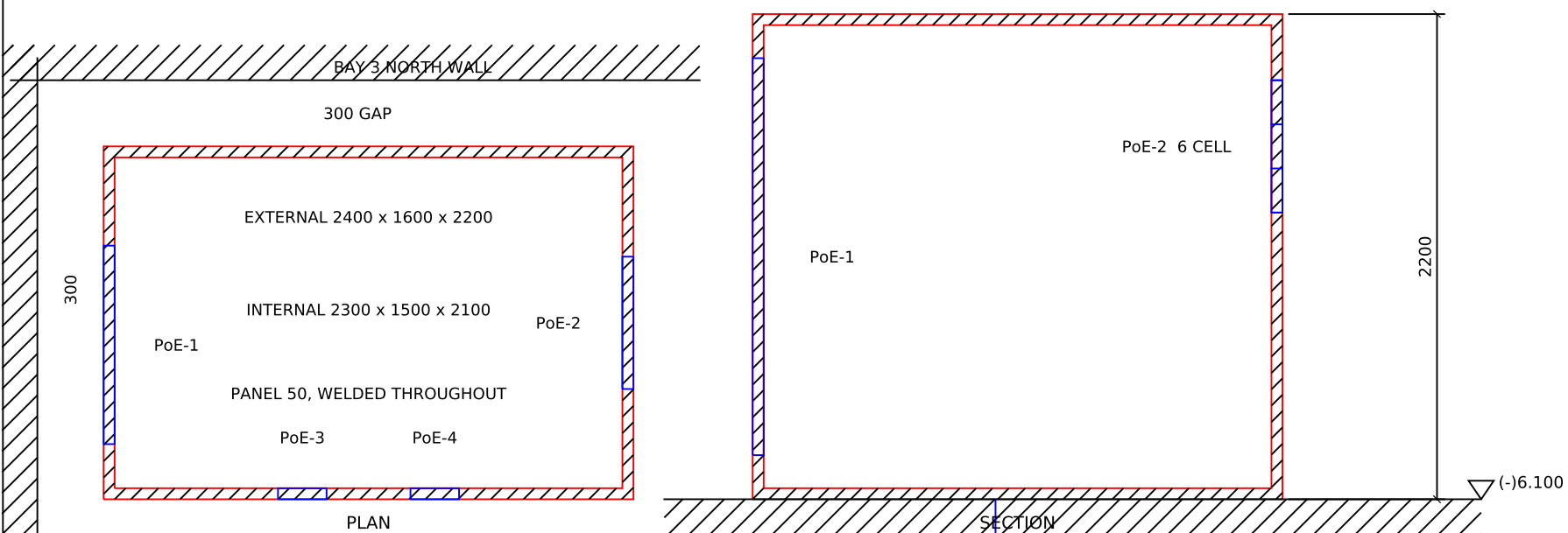




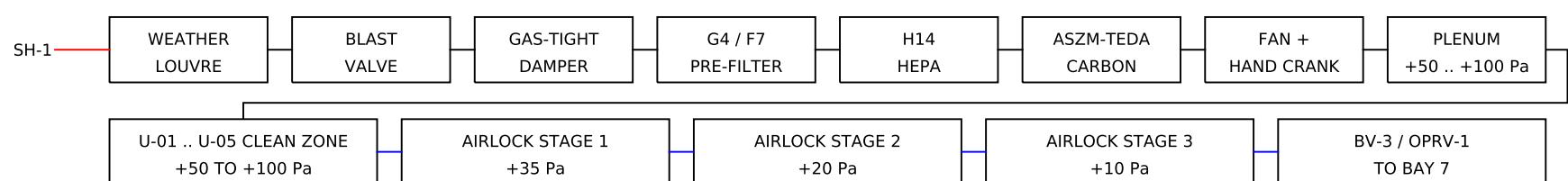
① HVAC SERVICES LAYOUT PLAN - UNDERGROUND LEVEL (-)6.100
1 : 100 FIVE BLAST VALVES ARE THE ONLY AIR PATHS THROUGH THE PROTECTIVE BOUNDARY



② EMP ZONE LAYOUT PLAN - UNDERGROUND LEVEL (-)6.100
1 : 100 EMP ZONE IS NOT THE CLEANLINESS ZONE



③ EMP ZONE 2 ENCLOSURE - PLAN AND SECTION
1 : 30 PANEL 50 WELDED THROUGHOUT



④ PROTECTIVE VENTILATION SCHEMATIC - FILTER TRAIN AND CASCADE
SCHEMATIC ONLY ONE TRAIN CARRIES THE WHOLE DUTY; THE PLENUM OVERPRESSURE SETS THE DIRECTION OF EVERY LEAK

HVAC EQUIPMENT SCHEDULE				
TAG	ITEM	DUTY	POSITION	NOTES
AHU-1	NBC FILTER TRAIN 1	300 m3/h	BAY 5	LOUVRE + BLAST VALVE + G4/F7 + H14 HEPA + ASZM-TEDA CARBON + FAN
AHU-2	NBC FILTER TRAIN 2	300 m3/h	BAY 5	IDENTICAL TO AHU-1, STANDBY. EITHER TRAIN CARRIES THE WHOLE DUTY
FAN-1	SUPPLY FAN, TRAIN 1	300 m3/h	WITHIN AHU-1	ELECTRIC DRIVE PLUS HAND CRANK
FAN-2	SUPPLY FAN, TRAIN 2	300 m3/h	WITHIN AHU-2	AS FAN-1
CO2-1	CO2 SCRUBBER, SODA LIME	20 kg/day	BAY 5	40 kg STORE = 48 h OF CLOSED MODE
O2-1	OXYGEN STORE	4.5 m3/day	BAY 5	2 No. 50 L CYLINDERS AT 150 bar = 15 m3
DH-1	DEHUMIDIFIER	-	BAY 5	CONDENSATE TO THE CLEAN SUMP THROUGH A 75 DEEP-SEAL TRAP
GEN-1	GENERATOR	15 kVA	BAY 8	COMBUSTION AND COOLING AIR 2600 m3/h THROUGH BV-4 / BV-5
SH-1	FRESH-AIR SHAFT	600 x 600	WEST OF THE BOX	GOOSENECK HEAD +1.500. 12.3 m FROM THE INTAKE TO THE ENTRY
SH-2	GENERATOR AIR SHAFT	600 x 600	X 22598 - 23198	SERVES BV-4 AND BV-5. OUTSIDE THE GAS-TIGHT ENVELOPE

BLAST VALVE AND GAS-TIGHT DAMPER SCHEDULE			
TAG	TYPE	POSITION	FUNCTION
BV-1	BLAST VALVE DN100	W1 WEST WALL, (598, 2200)	FRESH AIR, TRAIN 1
BV-2	BLAST VALVE DN100	W1 WEST WALL, (598, 4000)	FRESH AIR, TRAIN 2
BV-3	BLAST VALVE DN100	IN W6, (14998, 4900)	EXHAUST AND OVERPRESSURE RELIEF, TO BAY 7
BV-4	BLAST VALVE DN350	W4 EAST WALL, (21398, 1300)	GENERATOR INTAKE
BV-5	BLAST VALVE DN350	W4 EAST WALL, (21398, 4700)	GENERATOR EXHAUST
ALL FIVE: CLOSE IN UNDER 2 ms AND HOLD 1.3 s. RECESSED, IS 4991 CI. 6.2.1. DN100 DISC 3.0 kN; DN350 DISC 36.8 kN RECESSED			
GD-01	MANUAL GAS-TIGHT DAMPER	INBOARD OF BV-1	QUARTER-TURN, OPERABLE FROM INSIDE WITHOUT TOOLS
GD-02	MANUAL GAS-TIGHT DAMPER	INBOARD OF BV-2	AS GD-01
GD-03	MANUAL GAS-TIGHT DAMPER	INBOARD OF BV-3	AS GD-01
GD-04	MANUAL GAS-TIGHT DAMPER	INBOARD OF BV-4	AS GD-01
GD-05	MANUAL GAS-TIGHT DAMPER	INBOARD OF BV-5	AS GD-01
OPRV-1	OVERPRESSURE RELIEF VALVE	WITH BV-3, IN W6	HOLDS THE CLEAN ZONE AT +50 TO +100 Pa

EMP ZONE SCHEDULE			
ZONE	WHAT IT IS	SHIELDING MECHANISM	PERFORMANCE
EMP ZONE 0	EVERYTHING ABOVE GRADE	NONE CREDITED	0 dB
EMP ZONE 1	THE BURIED BOX, ALL EIGHT BAYS	REINFORCEMENT CAGE AT 150, BOTH CURTAINS	100 dB AT 10 kHz
EMP ZONE 2	WELDED STEEL ENCLOSURE, BAY 3	SOLID WELDED STEEL, PANEL 50	80 dB, 10 kHz - 1 GHz
EMP ZONE 2 EXTERNAL 2400 x 1600 x 2200 AT X 5820-8220, Y 3700-5300, INTERNAL 2300 x 1500 x 2100			
EMP ZONE 2 IS VERIFIED BY A FULL IEEE Std 299 SURVEY - A HOLD POINT BEFORE ANY EQUIPMENT IS INSTALLED			

ENVELOPE PENETRATION REGISTER				
TAG	KIND	BORE	WALL	HOST
BV-1	BLAST VALVE	100 DIA	600	W1 WEST WALL, BAY 1
BV-2	BLAST VALVE	100 DIA	600	W1 WEST WALL, BAY 1
BV-3	BLAST VALVE	100 DIA	400	W6, (14998, 4900)
BV-4	BLAST VALVE	350 DIA	600	EAST WALL, BAY 8
BV-5	BLAST VALVE	350 DIA	600	EAST WALL, BAY 8
SEP	SERVICE ENTRY PLATE	800 WIDE	600	NORTH WALL, X 11398-12198
PD-05	RISING MAIN BORE	50 DIA	600	THROUGH THE SERVICE ENTRY PLATE
ESC1	ESCAPE SHAFT	1400 DIA	3050	BAY 1, (2050, 2050)
ESC2	ESCAPE SHAFT	1400 DIA	3600	BAY 8, (19900, 2050)
VOID	STAIR VOID	3160 WIDE	900	PRESSURE SLAB, BAY 7

EMP ZONE 2 - POINT-OF-ENTRY SCHEDULE			
PoE	KIND	REF	TREATMENT
PoE-1	ACCESS - SHIELDED DOOR	5.4	RF-GASKETED OR KNIFE-EDGE SHIELDED DOOR, SE CERTIFIED TO THE ENCLOSURE
PoE-2	VENTILATION - HONEYCOMB	5.5	6 mm CELL x 25 mm DEEP WAVEGUIDE PANEL, FRAME WELDED TO THE ENCLOSURE
PoE-3	POWER - PCI	5.7.2.1	PROTECTIVE DEVICE ON EVERY CONDUCTOR, MOUNTED ON THE BOUNDARY
PoE-4	SIGNAL - FIBRE	5.7.4.1	OPTICAL FIBRE, NO METALLIC STRENGTH MEMBER AND NO METALLIC ARMOUR
PoE-5	BONDING AND EARTHING	IS 3043	100 x 50 x 3 STRAPS, 3.0 m x 16 mm RODS, EARTH RESISTANCE 5 ohm OR LESS

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS. WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH, STR AND SERVICES DRGS.
- PROJECT +X IS EAST AND PROJECT +Y IS NORTH. THE EXTERNAL WORKS ARE EAST OF THE BOX AND DOWNGRADIENT, AND STAND NORTH OF THE SENTRY POST ON THE OPPOSITE SIDE OF IT FROM THE APPROACH. THE FIVE BLAST VALVES ARE THE ONLY AIR PATHS THROUGH THE PROTECTIVE BOUNDARY. EACH IS RECESSED TO IS 4991 CI. 6.2.1 AND CARRIES A MANUAL GAS-TIGHT DAMPER INBOARD OF IT, OPERABLE FROM INSIDE WITHOUT TOOLS.
- FA-2 AND FA-3 CARRY RAW UNFILTERED AIR ACROSS THE CLEAN ZONE FROM BV-1 / BV-2 TO THE FILTER TRAINS. THEY ARE A PROTECTIVE ELEMENT, NOT A DUCT, AND ARE TO BE WELDED AND PRESSURE TESTED AS SUCH.
- EMP ZONE 2 IS DESIGNED TO 80 dB FROM 10 kHz TO 1 GHz TO MIL-STD-188-125-1, STANDING ALONE. NO ATTENUATION FROM THE CONCRETE BOX IS CREDITED AT ANY FREQUENCY.
- MAX BAR SPACING 150 IN BOTH CURTAINS IS AN EMP REQUIREMENT AND IS STRICTER THAN IS 456 CI. 26.3.3. EVERY CAST-IN FRAME AND SLEEVE IS WELDED TO THE REINFORCEMENT CAGE FOR SHIELD CONTINUITY.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

HVAC & EMP ZONE LAYOUT PLANS - UNDERGROUND LEVEL

CBRN HARDENED UG OPS ROOM

SHEET 10

Project number	01
Date	16 SEP 2026
Drawn by	SYN 01
Checked by	DR IR CHAUDHARI
MEP010	
Scale	As indicated